



Leveraging ChatGPT for personalized reflective learning in programming education: effects on self-efficacy, higher-order thinking, and project implementation skills

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Abstract

Reflective learning is crucial in programming education, fostering critical thinking and problem-solving skills. However, traditional guided reflection often falls short in addressing individual challenges and providing personalized feedback. Recent studies exploring large language models like ChatGPT for personalized feedback have primarily focused on immediate solutions, potentially hindering the development of higher-order thinking skills (HOTS). This study investigates the efficacy of ChatGPT-generated personalized feedback reflective reports on undergraduate programming students' self-efficacy, HOTS, and project implementation skills. Using a quasi-experimental design, 79 students in a Python programming course were divided into experimental and control groups over two semesters. The experimental group received ChatGPT-generated reports based on their error codes, while the control group engaged in traditional guided reflection. Results showed that the ChatGPT-facilitated approach significantly enhanced students' self-efficacy in logical thinking, algorithm design, and debugging. Moreover, students demonstrated improved critical thinking and problem-solving skills. The experimental group also produced more logical, useful, and well-crafted projects, as evaluated using the Creative Product Analysis Matrix (CPAM). This research underscores the potential of AI-supported personalized feedback in enhancing reflective learning practices and HOTS development in undergraduate programming education. It offers valuable insights for educators and researchers in higher education contexts, potentially transforming how universities prepare students for the challenges of the digital age and contributing to the development of a more skilled and adaptable workforce.

Keywords ChatGPT · Higher education · Higher-order thinking skills · Programming education · Project implementation skills · Reflective learning · Self-efficacy

1 Introduction

Programming has emerged as a crucial skill in today's technological landscape, playing a pivotal role in fostering critical thinking, problem-solving, and innovation across various disciplines in higher education (Lee et al., 2023b; Ouyang et al., 2022; Sun et al., 2021). As technological advancements continue to reshape industries and societal structures, the ability to understand and manipulate code has become increasingly essential for undergraduate students pursuing diverse careers (Weintrop et al., 2016). Programming not only serves as a fundamental skill for computer science majors but also acts as a powerful tool for modeling complex systems, analyzing data, and developing solutions in numerous fields, making it a vital component of modern higher education curricula (Lee, et al. 2023a; Sun et al., 2021).

Given this critical importance, it is essential to continually refine and enhance programming education methodologies at the undergraduate level. Two key factors significantly impact students' success in programming education: reflective learning and self-efficacy. Reflective learning, which involves analyzing and evaluating one's experiences to gain new insights, has been shown to enhance metacognition, critical thinking, and problem-solving abilities—essential components of higher-order thinking skills (HOTS) in higher education (Farahian et al., 2021; Lin et al., 2023). Complementarily, self-efficacy, or an individual's belief in their ability to succeed in specific tasks (Bandura, 1977), plays a significant role in undergraduate programming education (Tsai, 2019; Tsai et al., 2018). Research indicates that university students with higher self-efficacy in programming are more likely to persist in the face of challenges, engage in deeper learning strategies, and achieve better outcomes (Kovari & Katona, 2023; Yildiz Durak, 2018).

Traditional reflective methods in undergraduate programming courses often rely on guided reflection, where learners assess their coding errors and experiences based on predefined questions (Hug et al., 2021; Sun & Xu, 2023; Villareale et al., 2020). While this approach provides structure, it has several limitations in the higher education context. First, it may not adequately capture individual learners' unique challenges at the undergraduate level (Wu, 2023). Predefined questions often fail to align with specific difficulties faced by each learner, leading to superficial reflections (Sweet, 2022; Wu, 2023). Second, guided reflection relies on learners' ability to accurately identify and articulate their mistakes, which can be challenging for undergraduate students still developing their metacognitive skills (Chan & Lee, 2021). Third, this approach may not provide targeted feedback based on specific errors and misconceptions common in undergraduate programming courses (Qian & Lehman, 2019). While it encourages learners to think about their mistakes, it often lacks the personalized support needed to help them understand the underlying causes of their errors and develop effective strategies for improvement (Binks et al., 2022).

These limitations can be understood through the lens of several learning theories. The Cognitive Apprenticeship Model (Collins et al., 1991) emphasizes the importance of making thinking visible and providing contextualized feedback, which generic reflection prompts often fail to achieve. Similarly, Vygotsky's Zone of Proximal Development theory (Vygotsky & Cole, 1978) suggests that learning is optimized when support is tailored to an individual's current abilities and potential

growth areas, highlighting why standardized reflection questions may not effectively scaffold each learner's unique developmental needs. In programming education specifically, this theory is particularly relevant as students navigate the gap between their current understanding and the complex problem-solving required for effective coding (Lee et al., 2023b).

Furthermore, Scaffolding theory (Wood et al., 1976) illuminates this limitation by emphasizing the need for temporary, adaptive support that gradually fades as learners develop competence—a dynamic aspect that predetermined reflection prompts cannot provide. Self-Regulated Learning theory (Zimmerman, 2002) underscores the importance of personalized feedback in helping learners develop metacognitive awareness and self-regulatory strategies, which are essential for effective reflection but often underdeveloped in undergraduate programming students when using traditional guided reflection approaches.

Recent studies have explored using large language models (LLMs) like ChatGPT to provide personalized feedback in higher education programming courses (Lee et al., 2024; Wu et al., 2024). While these studies have shown the potential of ChatGPT in assisting with coding and problem-solving, they often focus on providing immediate solutions, which may lead to overreliance on AI and undermine critical thinking and problem-solving skills, essential components of HOTS in higher education (Ahmed & Sharo, 2023; Lee et al., 2024).

When considering AI-driven feedback systems in programming education, it is important to recognize the limitations of traditional intelligent tutoring systems (ITS). Conventional ITS approaches typically rely on predefined rule sets and structured knowledge bases to analyze student code and provide feedback (Keuning et al., 2018). While these systems have demonstrated value in certain educational contexts, they often lack the adaptability required to address the diverse and nuanced challenges faced by programming students (Lee et al., 2023a). Traditional ITS frequently prioritize identifying and correcting specific syntax or semantic errors rather than encouraging students to reflect deeply on their problem-solving processes and conceptual understanding (Rivers & Koedinger, 2017). This limitation aligns with the aforementioned challenges of traditional guided reflection, as both approaches may fail to provide the personalized scaffolding necessary for effective metacognitive development in programming education.

To address these concerns, this study proposes a novel approach to reflective learning in undergraduate programming courses. Instead of using ChatGPT to directly solve coding problems, this study leverages ChatGPT's language understanding capabilities to provide personalized feedback based on learners' error codes. By collecting students' error codes from classroom experiences and generating reflective reports with ChatGPT's assistance, this approach ensures students engage in independent thinking and problem-solving before receiving targeted feedback.

The integration of ChatGPT into the reflective learning process is theoretically grounded in several educational frameworks. Cognitive Load Theory (Sweller, 1988) suggests that effective learning occurs when instructional methods minimize extraneous cognitive load while optimizing germane cognitive load that contributes to schema acquisition. By providing targeted feedback based on students' specific errors, ChatGPT can help manage cognitive load more effectively than traditional

reflection methods. Additionally, the Formative Feedback framework (Shute, 2008) emphasizes the importance of timely, specific feedback aligned with learning goals, which the ChatGPT-generated personalized feedback aims to provide.

Problem-Based Learning theory (Barrows, 1996) supports this study of encouraging independent problem-solving before providing guidance, which this study's methodology implements by collecting error codes during students' independent work before generating reflective reports. Perhaps most relevantly, metacognitive scaffolding (Azevedo, 2005) provides a theoretical basis for how AI-generated feedback can support students' development of higher-order thinking skills. Unlike traditional scaffolding, metacognitive scaffolding specifically targets learners' ability to plan, monitor, and evaluate their own cognitive processes—skills that are crucial for programming success but challenging to develop through conventional reflection methods (Azevedo et al., 2022).

This study is deliberately designed to mitigate a significant concern with AI integration in education: the potential for over-reliance on AI-generated solutions, which could impede the development of independent problem-solving and self-regulation skills (Lee et al., 2024). By requiring students to first engage with programming challenges independently and only providing AI-assisted reflection after this engagement, the approach aligns with Self-Regulated Learning theory (Zimmerman, 2002), which emphasizes the importance of learners developing their own planning, monitoring, and evaluation capabilities. As Zimmerman (2002) notes, self-regulated learners are *“metacognitively, motivationally, and behaviorally active participants in their own learning process.”* The timing of ChatGPT-generated feedback in this study—after independent problem-solving attempts rather than during them—preserves the critical space needed for students to develop these self-regulatory skills while still benefiting from personalized guidance. This balance is essential, as previous research has cautioned that immediate AI assistance could potentially reduce students' cognitive effort and impede the development of critical reflection skills (Wu et al., 2024).

In designing this study, we carefully selected three specific dependent variables—self-efficacy, HOTS, and project implementation skills—based on both theoretical foundations and practical considerations in programming education. Self-efficacy was chosen because it strongly predicts persistence in challenging programming tasks and serves as a crucial mediator between instructional interventions and learning outcomes (Tsai, 2019). Unlike more transient measures such as motivation or engagement, self-efficacy represents a stable construct that influences long-term learning behaviors and career choices in programming fields (Yildiz Durak, 2018). Similarly, HOTS were selected over simpler knowledge retention metrics because programming education fundamentally aims to develop students' abilities to analyze complex problems, synthesize solutions, and evaluate code effectiveness—cognitive skills that extend beyond basic recall or comprehension (Huang et al., 2024). Finally, project implementation skills were included to assess the practical application of programming knowledge in authentic contexts, providing a more valid measure of real-world preparedness than traditional assessments (Lee et al., 2023b). Together, these three constructs offer complementary perspectives on students' cognitive development (HOTS), psychological readiness (self-efficacy), and practical capabilities

(project implementation)—a comprehensive framework for evaluating the effectiveness of educational interventions in programming.

The proposed reflective reports aim to offer deeper insights into coding mistakes and thought processes compared to traditional guided reflection methods in higher education settings. By integrating ChatGPT-based feedback into the reflection process after students have attempted to solve problems independently, this study seeks to enhance the effectiveness of reflection in promoting project implementation skills, HOTS, and self-efficacy among undergraduate programming students. To ensure internal validity, we conducted pre-study assessments of students' programming proficiency and prior knowledge, using these baselines as covariates in our analyses to control for individual differences that might affect responsiveness to the intervention. The research questions guiding this study are:

1. To what extent do ChatGPT-generated personalized feedback reflective reports influence learners' self-efficacy compared to traditional guided reflection?
2. What are the differences in learners' HOTS when provided with ChatGPT-generated personalized feedback versus traditional guided reflection?
3. How do ChatGPT-generated personalized feedback reflective reports impact learners' project implementation skills compared to traditional guided reflection?

Addressing these questions aims to enhance the learning outcomes and self-efficacy of undergraduate programming students through innovative reflective learning approaches leveraging LLMs like ChatGPT. This study aims to demonstrate the potential of ChatGPT-generated personalized feedback reflective reports in transforming how undergraduate programming students engage in reflective learning and develop their skills, ultimately contributing to the advancement of programming education in higher education and preparing students for the technological demands of the future.

2 Related work

2.1 Reflection in programming education

Reflection plays a crucial role in programming education, enabling students to analyze their learning experiences, identify strengths and weaknesses, and develop improvement strategies (Qian & Lehman, 2017). Studies highlight the importance of reflection in promoting deep learning, metacognition, and self-regulation (Vosniadou et al., 2021; Zarestky et al., 2022). Reflective activities, such as self-explanation and peer review, have been shown to enhance students' understanding of programming concepts, problem-solving skills, and critical thinking abilities (Alt & Raichel, 2022; Huang et al., 2024). Additionally, reflective learning positively impacts students' self-efficacy and motivation in programming (Hsia & Hwang, 2020; Lin et al., 2021).

Despite the recognized importance of reflection, traditional methods, such as guided reflection, often face limitations (Wu, 2023). Guided reflection may not adequately capture individual learners' unique challenges and misconceptions, leading

to superficial reflections (Sweet, 2022; Wu, 2023). Furthermore, it often lacks personalized feedback based on specific errors, hindering deep, transformative reflection (Qian & Lehman, 2019).

The Cognitive Apprenticeship Model (Collins et al., 1991), introduced earlier, emphasizes making thinking processes visible through modeling, coaching, and scaffolding—elements often absent in traditional guided reflection approaches. In programming education, this model has been applied in pair programming settings where experts demonstrate thinking processes while coding (Denner et al., 2014), but rarely in structured reflection activities. Similarly, Vygotsky's Zone of Proximal Development theory (Vygotsky & Cole, 1978) has informed some adaptive learning systems that adjust difficulty based on learners' abilities (Wu et al., 2023b), yet these principles have not been widely applied to personalize reflection in programming education specifically. Scaffolding theory (Wood et al., 1976) offers further insights into how temporary, adaptive support can guide learners toward independent problem-solving. While this theory has informed various programming education approaches (Grover et al., 2015; Lee et al., 2023b), its application to reflective learning in programming contexts remains underexplored, particularly regarding how automated systems might provide scaffolding that adapts to individual learners' reflective needs.

Recent studies have explored learning analytics and intelligent tutoring systems to support reflective learning in programming education (López-Pernas et al., 2021; Paladines & Ramirez, 2020). However, these approaches often rely on predefined algorithms and rule-based systems, which may not fully capture the complexity and diversity of students' learning experiences (Jaiswal et al., 2021; Liu et al., 2019). Moreover, feedback from these systems may lack the natural language fluency and contextual understanding necessary for effective reflective learning support (Paladines & Ramirez, 2020).

To address these limitations, this study proposes a novel approach to reflective learning in programming education, leveraging the LLMs, ChatGPT to generate personalized reflective reports based on learners' error codes. By utilizing ChatGPT's natural language processing capabilities and its ability to understand and generate human-like text, this approach aims to provide targeted, contextually relevant, and pedagogically sound feedback tailored to each student's specific needs and misconceptions. This innovative method addresses the shortcomings of traditional guided reflection and existing learning analytics and intelligent tutoring systems, offering a more effective and personalized reflective learning experience that enhances HOTS and self-efficacy in programming education.

2.2 Higher-order thinking skills

Higher-order thinking skills (HOTS), such as critical thinking, problem-solving, and creativity, are essential for success in programming education and beyond (Li et al., 2023). These skills enable learners to analyze complex problems, generate innovative solutions, and adapt to novel challenges (Farahian et al., 2021; Lin et al., 2023). In programming, HOTS are crucial for designing efficient algorithms, debugging com-

plex code, and developing original software solutions (Cheng et al., 2023; Lee et al., 2023b).

Numerous studies have investigated the development of HOTS in programming education. For example, Anggraeni et al. (2023) examined the effects of problem-based learning on students' critical thinking and problem-solving skills. They found that students engaged in problem-based learning showed significant improvements in analyzing problems, generating solutions, and evaluating their work. Similarly, Xu et al. (2023) explored the impact of collaborative learning methods, such as pair programming, on students' creativity and metacognition. Their findings suggest that collaborative learning enhances students' ability to generate novel ideas, provide constructive feedback, and reflect on their learning processes.

Reflective learning activities also promote HOTS in programming education. Tamang et al. (2021) found that self-explanation prompts in a programming course improved students' problem-solving skills and understanding of programming concepts. Pham et al. (2020) reported that reflective learning activities, such as journaling and peer feedback, significantly enhanced students' critical thinking skills.

Self-Regulated Learning theory (Zimmerman, 2002), as mentioned in the introduction, underpins many approaches to developing HOTS in programming education. Research by Wang et al. (2024b) demonstrated that explicit instruction in self-regulation strategies improved students' problem-solving abilities and debugging skills in programming courses. Similarly, Cognitive Load Theory (Sweller, 1988) has informed instructional design approaches that manage complexity for novice programmers (Wang et al., 2024a), showing that reducing extraneous cognitive load during problem-solving activities allows students to dedicate more mental resources to developing higher-order thinking skills (Huang et al., 2025). Metacognitive scaffolding theory (Azevedo, 2005) extends these frameworks by specifically addressing how external support can enhance learners' metacognitive capabilities—including planning, monitoring, and evaluating their own thinking processes. This theoretical perspective is particularly relevant for programming education, where metacognitive skills strongly predict success in complex problem-solving tasks (Kim & Lim, 2019). Recent studies have begun to explore how metacognitive scaffolding can be implemented through technological tools (Azevedo et al., 2022; Kim & Lim, 2019), but its application through AI-generated personalized feedback remains an emerging area that this study aims to address.

Despite the growing body of research on HOTS in programming education, there is still a need for innovative approaches that provide personalized support and guidance to learners. Existing strategies, such as problem-based and collaborative learning, often rely on generic prompts and activities that may not fully address individual learners' specific challenges and misconceptions (Choo, 2012; Scheibenzuber et al., 2021). Reflective learning activities, while promising, often lack the targeted feedback necessary to facilitate deep reflection and skill development.

To address these limitations, this study proposes a novel reflective learning approach leveraging ChatGPT-generated personalized feedback to promote HOTS in programming education. By providing targeted guidance based on learners' specific misconceptions and error patterns, this approach aims to facilitate deeper reflection and foster the development of critical thinking, problem-solving, and creativity in

programming students. This innovative approach extends existing research on HOTS by offering a more personalized and adaptive learning experience that addresses individual learners' unique needs and challenges.

2.3 ChatGPT in programming education

The integration of LLMs, such as ChatGPT, in programming education has gained significant attention in recent years (Phung et al., 2023; Yilmaz & Karaoglan Yilmaz, 2023). ChatGPT's ability to understand and generate human-like text has led researchers to explore its potential in assisting students with programming tasks, such as problem-solving and code generation (Hoq et al., 2024; Joshi et al., 2024). For instance, Fenu et al. (2024) developed a ChatGPT-based tutoring system that provided personalized feedback to students learning programming, resulting in improved problem-solving skills and engagement.

This growing interest in AI-based systems extends beyond ChatGPT to encompass a range of adaptive learning approaches in programming education. Notably, the integration of AI-driven adaptive learning with flipped classroom models has shown particular promise. Gasaymeh (2024) demonstrated that coupling AI-generated personalised feedback with a flipped structure significantly boosted student achievement in introductory programming courses. Likewise, Kuo and Chang (2023) embedded an Apriori-based learning-diagnosis engine in a flipped programming classroom: pre-class quizzes were mined for concept-level association rules, automatically generating personalised concept maps and forming adaptive peer groups. This data-driven grouping led to statistically significant improvements in learning effectiveness, motivation, self-efficacy, reduced cognitive load, and more positive programming attitudes when compared with conventional, non-adaptive flipped classrooms. Together, these studies highlight how AI-enabled adaptivity can amplify the pedagogical advantages of the flipped model in programming education.

While these studies demonstrate the potential benefits of ChatGPT in programming education, concerns have been raised about potential overreliance on ChatGPT-generated solutions and their impact on students' learning outcomes (Lee et al., 2024). Directly using ChatGPT to solve programming problems may tempt students to accept generated solutions without actively engaging in the problem-solving process, hindering the development of essential programming skills and limiting knowledge transfer (Ahmed & Sharo, 2023). These concerns are particularly significant when viewed through the lens of Self-Regulated Learning theory (Zimmerman, 2002), which emphasizes the importance of learners developing their own planning, monitoring, and self-evaluation capabilities. When students rely too heavily on AI-generated solutions, they bypass the critical self-regulatory processes that Zimmerman identifies as essential for meaningful learning and skill development. This can potentially undermine the development of metacognitive skills necessary for programming proficiency and lifelong learning.

Recent research has begun to explore how the integration of ChatGPT aligns with established educational frameworks (Li et al., 2024; Wu et al., 2025). Some studies have examined ChatGPT's potential through the lens of Formative Feedback (Huang et al., 2025; Shute, 2008), showing that AI-generated feedback can provide timely

guidance that helps students identify gaps in their understanding (Wu et al., 2024). Building on Problem-Based Learning theory (Barrows, 1996), Lee et al., (2024) found that when ChatGPT was used to support—rather than replace—students’ problem-solving processes, it enhanced their engagement with challenging programming tasks and improved learning outcomes. From a metacognitive scaffolding perspective (Azevedo, 2005), AI-generated feedback may serve as an external regulatory agent that supports students’ development of self-monitoring and self-evaluation skills (Lee et al., 2024). However, research specifically examining how LLMs like ChatGPT can provide metacognitive scaffolding in programming education is still in its nascent stages, representing a significant gap that this study aims to address.

A comprehensive overview of AI-based adaptive learning in programming education reveals several distinct approaches beyond just ChatGPT integration. Keuning et al. (2018) conducted a systematic literature review that categorized automated feedback systems into knowledge-based approaches that leverage domain expertise, data-driven techniques that learn from student interactions, and hybrid systems that combine multiple strategies. Significantly, recent research by Lee et al. (2023a) demonstrated that machine learning algorithms utilizing behavioral data can predict programming difficulties and proactively provide personalized support before students encounter frustration points. Meanwhile, Marwan et al. (2022) developed an adaptive system that progressively adjusts programming task complexity based on real-time performance analysis, demonstrating improved learning trajectories compared to static sequencing approaches.

Student perceptions of AI-enhanced adaptive learning systems for programming education warrant particular attention, as they significantly influence adoption and effectiveness. Research by Liu et al. (2023) found that while students generally appreciate personalized feedback from AI systems, their acceptance is strongly mediated by transparency—students showed greater trust and engagement when they understood how the AI generated its recommendations. Additionally, Huang et al. (2025) reported that students’ perceptions evolved over time, with initial skepticism giving way to appreciation as they experienced the benefits of personalized guidance. These findings underscore the importance of thoughtful implementation strategies that address student concerns and build trust in AI-enhanced learning environments.

To address these concerns, researchers have investigated innovative approaches to integrate ChatGPT into programming education while promoting active learning and HOTS (Al Shloul et al., 2024; Lee et al., 2024; Tsai, 2023). One promising approach is providing reflective reports that give students time to think and attempt problem-solving independently before receiving targeted feedback. Lee et al. (2024) proposed a system leveraging ChatGPT to generate personalized feedback based on students’ question, encouraging development of problem-solving strategies before receiving ChatGPT-generated guidance. Similarly, Ouh et al. (2023) investigated using ChatGPT-generated code explanations as a basis for reflective learning activities, promoting critical thinking and deeper understanding of programming concepts.

Despite these promising findings, existing research on ChatGPT in programming education has primarily focused on its direct application in problem-solving and code generation. More research is needed on effectively integrating ChatGPT into reflec-

tive learning processes to promote HOTS and self-efficacy while mitigating the risks of overreliance on AI-generated solutions.

The evolution of AI-driven feedback systems in programming education provides important context for understanding the potential contribution of the current study. Early intelligent tutoring systems for programming, such as PROUST (Johnson & Soloway, 1985) and LISP-TUTOR (Anderson et al., 1989), pioneered automated feedback but were constrained by domain-specific knowledge bases and limited programming languages. These systems established the foundation for automated programming guidance but lacked the flexibility to address diverse student needs and programming approaches (Crow et al., 2018).

As the field progressed, more sophisticated systems emerged that attempted to provide adaptive feedback through data-driven approaches. Systems such as the Adaptive Immediate Feedback (AIF) system developed by Marwan et al. (2022), and the data-driven intervention framework proposed by Lee et al. (2023a), incorporated deep learning techniques to analyze student behavior and offer personalized guidance. These systems leveraged real-time interaction data or historical traces to present context-aware feedback, improving student persistence and task completion. However, these advances still faced challenges in promoting deeper student reflection, often focusing more on immediate correctness than on conceptual understanding.

Contemporary systems have become increasingly sophisticated but continue to face limitations in supporting reflective learning. For example, although metacognitive scaffolding has been integrated into intelligent tutoring systems, many tools still struggle to help students develop metacognitive regulation during programming tasks. For instance, the Archimedes system emphasizes cognitive modeling but only partially supports reflective feedback and self-assessment strategies tailored to learners' cognitive pathways (Beck & Mohammadi-Aragh, 2020). Similarly, although metacognitive supports like reflective prompts and graphical organizers are known to be effective, their integration remains limited in current programming environments, often failing to adapt to individual students' evolving understanding (Rum & Ismail, 2017).

These limitations in existing systems highlight a critical gap that aligns with the challenges of traditional guided reflection discussed earlier: the need for AI-driven feedback mechanisms that transcend error correction to foster metacognitive reflection, self-efficacy, and HOTS. The emergence of LLMs like ChatGPT presents an opportunity to address this gap through their advanced language understanding capabilities and contextual awareness (Wu et al., 2025).

The approach proposed in this study differs from previous intelligent tutoring systems in several substantive ways that address the limitations of both traditional guided reflection and conventional AI-driven feedback. Unlike rule-based systems constrained by predefined knowledge structures, the ChatGPT-based approach leverages contextual understanding to generate nuanced feedback tailored to individual learners' specific error patterns and misconceptions. Furthermore, where many existing systems focus primarily on technical correction, our approach emphasizes the development of reflective thinking by encouraging students to analyze their errors within a broader context of programming concepts and problem-solving strategies. This integration of automated error analysis with personalized reflective guidance

offers a more comprehensive learning experience that aligns with established theories of reflective learning and metacognitive development discussed earlier.

3 ChatGPT-generated personalized feedback reflective reports

3.1 Reflective report system in jupyter notebook

The reflective report system was developed using Jupyter Notebook, an interactive development environment for Python programming. This platform was selected for its intuitive interface that supports both code execution and markdown documentation, creating an integrated environment for instructional content and coding practice (Fig. 1).

To enable personalized reflective reports, the system utilizes the nbextensions package to extend Jupyter Notebook's functionality. A custom extension automatically logs students' executed code and results to a Firebase database in real-time. The system workflow (Fig. 2) consists of three key components:

First, the user authentication component requires students to enter their unique identifier and name upon accessing the system (Fig. 3). This information is stored in the Firebase database, enabling the association of each student's activities with their individual profile.

Second, the data capture component automatically logs students' coding activities, including executed code, results, and encountered errors. This real-time data collection provides a comprehensive record of each student's learning journey, capturing error information based on Python's built-in error categories.

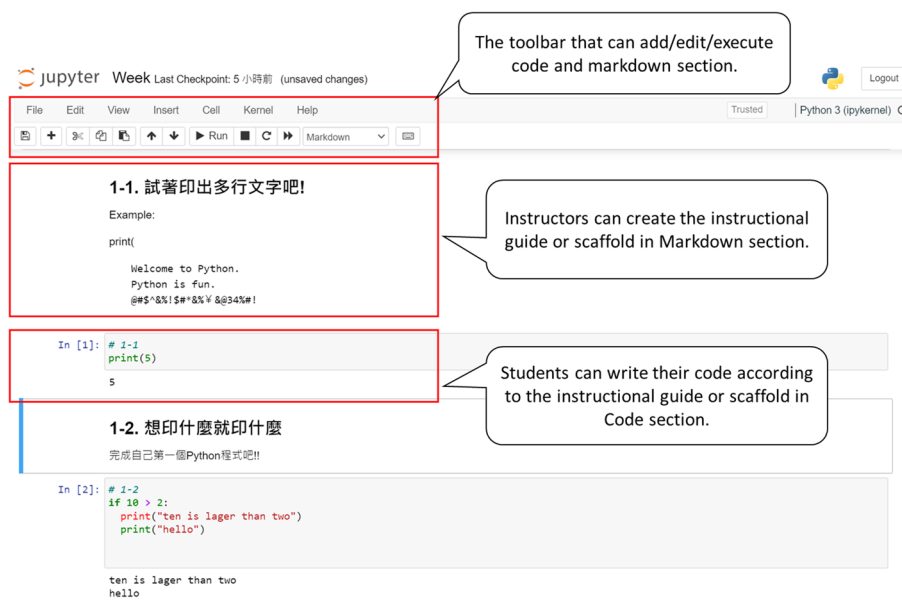


Fig. 1 Interface of original Jupyter notebook

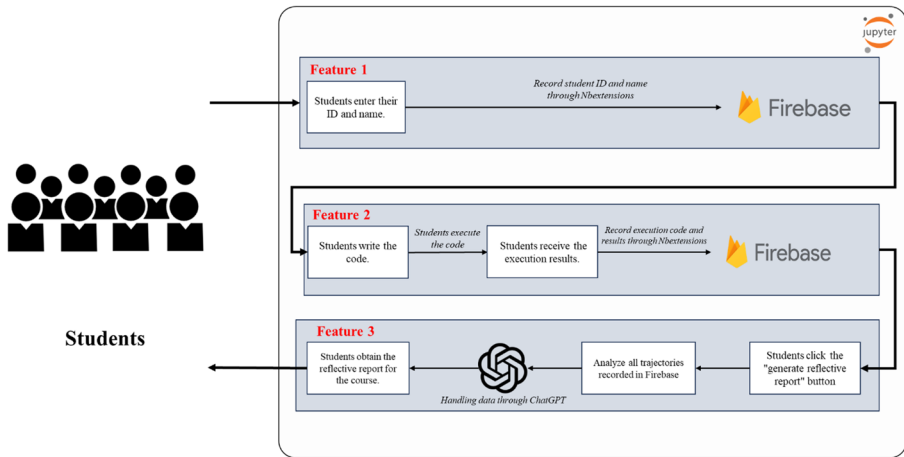


Fig. 2 The workflow of reflective report system

The screenshot shows the Jupyter interface for entering student ID and name. The interface includes a menu bar (File, Edit, View, Insert, Cell, Kernel, Help) and a toolbar with various icons. Below the toolbar, there are two input fields: "姓名:" (Name) and "學號:" (Student ID). A green button labeled "送出" (Send) is located to the right of the input fields. The interface also displays "Task1 Last Checkpoint: 16 小時前 (autosaved)" and a "Logout" button.

Fig. 3 Interface of entering student ID and name

Third, the report generation component allows students to request a personalized report by clicking the "Generate Reflective Report" button (Fig. 4) upon completing a programming lesson.

This action triggers ChatGPT to analyze the student's logged error codes and generate a tailored reflective report. The personalized report (Fig. 5) presents a breakdown of the student's error categories along with customized hints and guidance for addressing each type of error.

The integration of ChatGPT into the reflective report system provides several advantages over traditional static feedback mechanisms. ChatGPT's ability to generate coherent, contextually relevant, and understandable feedback tailored to each student's unique needs enhances the effectiveness of the reflective learning process. As the system continues to interact with students, the quality and specificity of its feedback can improve over time, creating a dynamic and evolving learning support system.

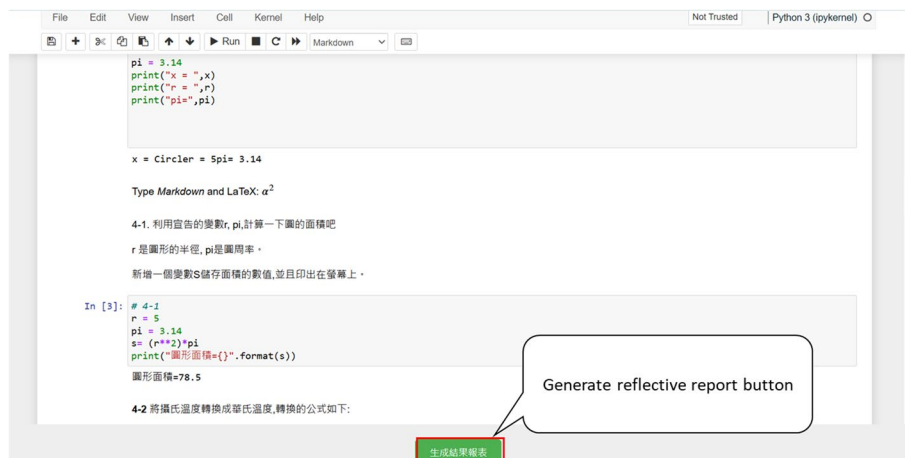


Fig. 4 Interface of Generate reflective report button

錯誤類型	出現次數	改進建議
NameError	1	希望這些建議對您有所幫助,記得在寫程式時要注意變數命名的準確性,這樣可以避免像'NameError'這樣的錯誤產生。另外,也記得熟悉基本的數學運算符號及函數的使用,這能幫助您更有效率地撰寫程式。若有任何疑問或需要進一步解說,請隨時向我提問!
IndentationError	3	在程式撰寫過程中,縮排是非常重要的,它不僅可以提升程式碼的可讀性,更可以避免出現像是IndentationError這樣的錯誤。請在撰寫程式時特別留意程式碼的結構,確保所有的程式區塊都有正確的縮排。另外,建議您在撰寫程式時可以多加練習,並尋求他人的幫助或反饋。隨著實際經驗的累積,你將能夠更加熟練地撰寫程式碼並避免一些常見的錯誤。加油!如果有任何程式碼相關的問題或疑問,歡迎隨時向我尋求幫助。
TypeError	1	對於你的錯誤描述,感謝你提供了清楚的問題描述,這對於找出程式碼中的問題是非常有助的,以下是一些建議和提醒以幫助你更好地寫程式: 1. 在 Python 中,確保變數的資料型態是正確的非常重要,尤其是在進行數學運算時,確保參與運算的變數是相容的資料型態,這樣可以避免出現錯誤。 2. 在程式碼撰寫過程中,可以透過使用 'type()' 函數來檢查變數的資料型態,以確認它們是預期的型態。 3. 如果遇到 TypeError 錯誤,請先檢查參與運算的變數是否為正確的型態,並進行必要的轉換。 4. 當程式碼中多次使用到相同的數值或變數時,可以考慮將其定義為變數,這樣可以提高程式碼的可讀性和維護性。希望以上建議能幫助你避免類似的錯誤,並在寫程式時更加順利與準確。若有任何疑問或需要進一步的幫助,請隨時提問,加油!
SyntaxError	12	在寫程式時,經常會遇到各種各樣的錯誤,這是很正常的,關鍵在於如何從錯誤中學習並修正它們,以下是一些建議,可以幫助你避免這些常見錯誤,並提高您的程式設計技能: 1. 注意語法: 確保您的程式碼符合Python的語法規則,像是括號、引號、冒號等都必须有對應且正確使用。 2. 細心謹慎: 檢查程式碼時要仔細,特別是要注意引號、括號等的配對以及大小寫字母的使用。 3. 測試驗證: 在執行程式之前,最好先測試並驗證程式的每一個步驟,以確保程式邏輯正確。 4. 學習資源: 如果對於特定的語法或程式碼片段不熟悉,可以參考Python語言的官方文件或其他線上資源來學習補充知識。 5. 不斷實踐: 多練習寫程式,透過不斷的實踐來熟悉Python的語法和規則,進而提高自己的編碼水準。希望這些建議對您有所幫助,並能幫助您在未來的程式設計過程中更加順利地避免和解決類似的錯誤。如果有任何問題或需要進一步的幫助,請隨時向我提問!

Fig. 5 Interface of the reflective report

3.2 ChatGPT integration and design

The integration of ChatGPT into the reflective report system was carefully designed to ensure pedagogically sound feedback. The design choices were guided by established educational theories and frameworks.

Key hyperparameters were fine-tuned based on Cognitive Load Theory (Sweller, 1988) and the Goldilocks Principle (Metcalfe & Kornell, 2005):

- **Temperature:** A moderate value of 0.7 was selected to balance focused feedback with creative expression, aligning with Cognitive Load Theory by providing clear information while maintaining student engagement.
- **Frequency Penalty:** A value of 0.5 was chosen to ensure diverse responses without excessive repetition, adhering to the Goldilocks Principle by maintaining an appropriate level of variety while ensuring clarity and consistency.

Prompt engineering is vital in guiding ChatGPT to generate pedagogically sound feedback aligned with learning objectives. The prompts were crafted based on Formative Assessment (Black & Wiliam, 2009) and Self-Regulated Learning (Zimmerman, 2002) principles, including the following key elements:

Prompt engineering was based on Formative Feedback (Shute, 2008) and Self-Regulated Learning (Zimmerman, 2002) principles, including:

- **Error Context:** Including information about specific error types and frequencies to provide necessary context for targeted feedback (Shute, 2008).
- **Feedback Objectives:** Explicitly stating objectives to guide ChatGPT in focusing on explanations, misconceptions, and actionable improvement suggestions (Zimmerman, 2002).
- **Tone and Style:** Specifying an encouraging and supportive tone to foster a growth mindset.
- **Pedagogical Principles:** Incorporating best practices for effective feedback, such as focusing on specific behaviors and offering concrete examples, grounded in Formative Feedback (Shute, 2008).

The specific prompt structure used in this study is as follows:

“You are an AI tutor assisting a student in a Python programming course. The student has encountered the following errors while coding:

[List of error types and frequencies].

Please provide a reflective report for the student that includes:

1. *A brief explanation of each error type and its implications.*
2. *Identification of potential misconceptions or knowledge gaps based on these errors.*
3. *Actionable suggestions for improvement, including specific coding practices or concepts to focus on.*
4. *Encouragement for the student’s progress and motivation for continued learning.*

Your feedback should be:

- *Clear and easy to understand for a beginner programmer.*
- *Encouraging and supportive, fostering a growth mindset.*
- *Specific to the errors encountered, avoiding generic advice.*
- *Focused on promoting self-reflection and independent problem-solving.*

Limit your response to approximately 500 words.”

By designing prompts and fine-tuning ChatGPT’s parameters based on established educational theories, the system ensures pedagogically sound and effective feedback. This theory-driven approach maximizes the impact of personalized reflective reports in promoting deep learning, self-awareness, and skill improvement in programming education.

4 Methodology

4.1 Participants

This two-semester study was conducted at a university located in the southern region of Taiwan, focusing on a Python programming course. The research involved a total of 79 undergraduate students, who were divided into two distinct groups: the Control Group (CG), consisting of 39 participants, and the Experimental Group (EG), comprising 40 participants. Group assignment was based on enrollment semester rather than randomization, with the CG participating in the first semester and the EG in the second semester. This semester-based allocation was chosen to minimize cross-contamination between groups and maintain the ecological validity of the classroom environment.

Although the participants were not explicitly informed about their specific group allocation, they willingly took part in the experiment and data collection process after providing informed consent. The participants were fully aware of the study’s general objectives and nature. To minimize potential biases arising from non-randomized group assignment, this study implemented several control measures. First, this study ensured that both groups were subjected to identical teaching standards, including the allocation of the same instructor, the use of a consistent teaching syllabus, and the provision of uniform teaching materials and environments.

The baseline comparability of the groups was verified through pre-test assessments of self-efficacy, HOTS, and prior programming knowledge, with no significant differences found between groups at baseline ($p > 0.05$ for all measures). Additionally, demographic factors such as age distribution ($M_{EG} = 20.3$, $SD_{EG} = 1.2$; $M_{CG} = 20.5$, $SD_{CG} = 1.3$), gender ratio (EG: 62.5% male; CG: 64.1% male), and prior programming experience ($M_{EG} = 1.2$ years, $SD_{EG} = 0.8$; $M_{CG} = 1.1$ years, $SD_{CG} = 0.7$) were comparable between groups, further minimizing potential confounding variables.

The primary distinction between the EG and CG lay in their approach to reflective learning. The EG received reflective reports generated by the ChatGPT-based personalized feedback system, which leveraged students’ error codes to provide tar-

geted insights and recommendations for improvement. Conversely, the CG engaged in traditional guided reflection activities, which relied on predefined questions and prompts for students to assess their coding experiences and errors, without the personalized feedback offered by the ChatGPT-based system.

This study acknowledges that semester-based allocation rather than randomization represents a limitation of the study design. To account for potential semester effects (such as academic calendar differences or external events), we monitored and documented any significant semester-specific events that might influence learning outcomes and found no substantial differences in the educational context between semesters. The use of ANCOVA in our data analysis further helped control for any pre-existing differences between groups.

Throughout the study, the highest standards of research ethics were maintained, adhering to the principles outlined in the Declaration of Helsinki. The privacy of the participants was protected, and all collected data were anonymized. The Institutional Review Board for Human Subject Research at the university granted approval for the research protocol, ensuring that the study was conducted in an ethical manner.

4.2 Experimental procedure

This study employed a quasi-experimental design to investigate the effectiveness of ChatGPT-generated personalized feedback reflective reports on self-efficacy, HOTS, and project implementation skills. The study was conducted over two semesters, with the Control Group (CG) participating in the first semester and the Experimental Group (EG) in the second semester. The course focused on integrating Python programming with Raspberry Pi embedded development boards, combining software and hardware knowledge to solve real-world problems in IoT and AI domains.

As illustrated in Fig. 6, the experimental procedure consisted of a ten-week course structured into three weekly phases: basic knowledge instruction, project implementation, and reflection.

During the basic knowledge instruction phase, teachers helped students develop lower-order thinking skills by clarifying concepts and providing explanatory examples. The instruction was strictly standardized between semesters, using identical lesson plans, examples, and teaching materials to ensure consistency of the learning experience between groups. In the project implementation phase, students applied their knowledge to achieve project objectives, developing analytical and problem-solving abilities. Both groups received the same project assignments, resources, and technical support during this phase.

The key difference between groups occurred in the reflection phase:

- **Experimental Group (EG):** Students received ChatGPT-generated personalized feedback reflective reports based on their logged error codes. These reports included error categorization, frequency analysis, misconception explanations, and targeted improvement suggestions. Students used these reports to guide their reflection process and complete a reflective learning sheet.
- **Control Group (CG):** Students engaged in traditional guided reflection activities using pre-set questions and relying on their own recollection of challenges and

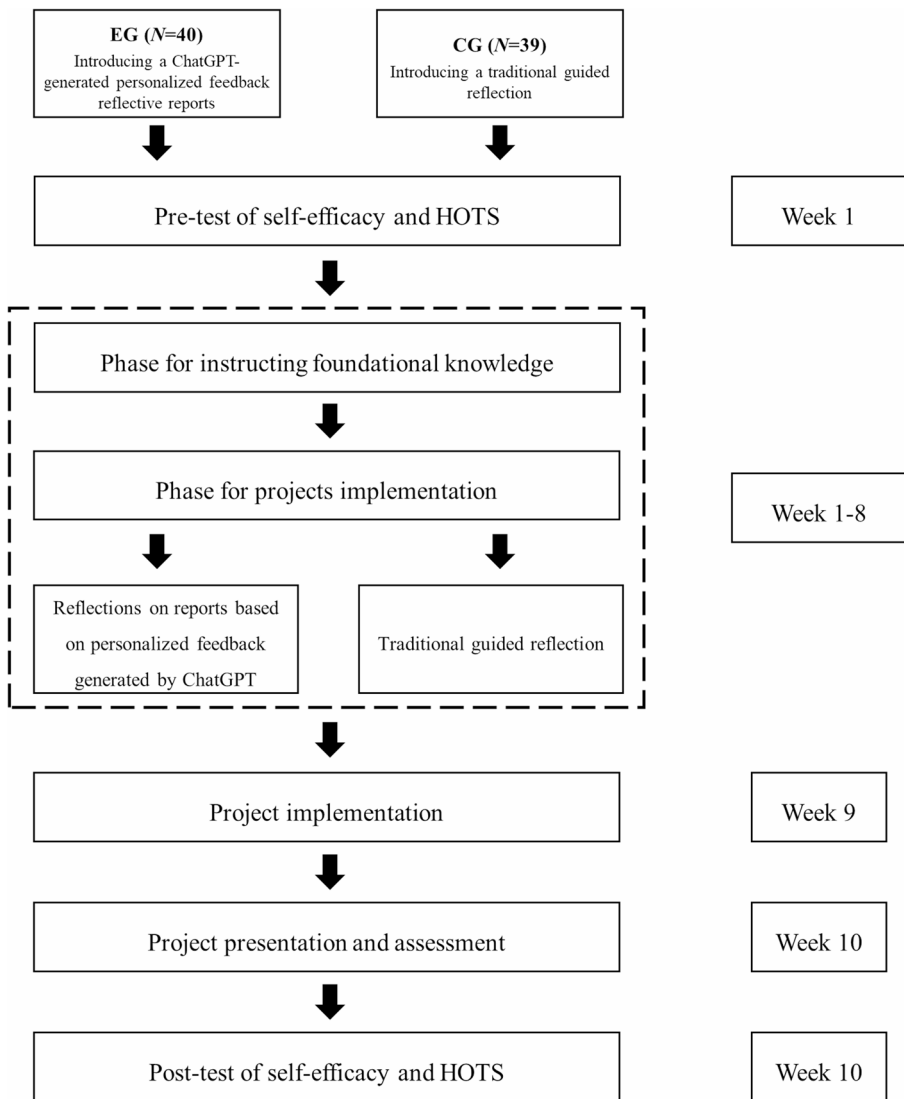


Fig. 6 Experimental procedure

errors. Without automated error analysis, these students completed their reflective learning sheets based on memory and self-analysis.

To ensure experimental equivalence beyond the intervention itself, both groups had comparable access to additional learning support. Both the EG and CG had equal access to instructor consultation during designated office hours (two hours per week), and both groups participated in the same peer discussion activities during regular class sessions. Neither group received specialized human tutoring beyond the standard instructor support available to all students. The instructor was careful to main-

tain consistent engagement with both groups, spending equivalent time addressing questions and providing general guidance. All project work was conducted individually, not in collaborative teams, further controlling for potential confounding variables related to peer influence.

The primary differences in the reflection process were the level of personalization, depth of analysis, and basis for reflection. The ChatGPT-generated reports provided highly personalized, data-driven insights for more targeted reflection, while the traditional method relied more heavily on students' ability to recall and analyze their learning experiences. Both approaches, however, required the same amount of time investment from students (approximately 30 min per reflection session), and both groups completed the same number of reflection activities throughout the course.

In the ninth week, students developed a project combining software and hardware knowledge to create a solution beneficial to human life. The tenth week was dedicated to project presentations and grading, with two experts evaluating the projects using the Creative Product Analysis Matrix (CPAM) coding scheme (Besemer & O'Quin, 1999). These experts were blind to group assignment, reducing potential bias in the evaluation process.

Both groups completed pre-tests and post-tests to measure changes in self-efficacy, HOTS, and project implementation skills. Pre-tests were conducted at the beginning of the first week and post-tests after the end of the tenth week, allowing for comprehensive assessment of the impact of ChatGPT-generated personalized feedback reflective reports. Additionally, we conducted pre-study assessments of students' programming proficiency to establish baseline equivalence between groups and incorporate these measures as covariates in subsequent analyses.

4.3 Research tools

4.3.1 Self-efficacy scale

To assess participants' self-efficacy, this study employed a scale adapted from the Computer Programming Self-Efficacy Scale developed by Tsai et al. (2018). This scale, designed for computer Literacy education, has demonstrated good reliability and validity in previous research. It consists of 16 items measuring five dimensions of self-efficacy: "Logical Thinking" (4 items), "Cooperation" (3 items), "Algorithm" (3 items), "Control" (3 items), and "Debug" (3 items). Items are rated on a five-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

For this study, the scale was translated into Chinese using the back-translation method (Brislin, 1970). First, a bilingual expert translated the original English scale into Chinese. Then, another bilingual expert, who had not seen the original version, translated the Chinese version back into English. The two English versions were compared, and any discrepancies were resolved through discussion and revision of the Chinese translation.

To assess the reliability of the translated scale, a pilot study was conducted with 30 undergraduate students not part of the main study. Cronbach's alpha coefficients were calculated for the overall scale and its five dimensions. The overall scale exhibited high internal consistency ($\alpha=0.88$), as did the "Logical Thinking" ($\alpha=0.84$),

“Cooperation” ($\alpha=0.86$), “Algorithm” ($\alpha=0.85$), “Control” ($\alpha=0.87$), and “Debug” ($\alpha=0.83$) dimensions, indicating sufficient reliability for use in this study (Nunnally, 1978).

The self-efficacy scale was administered to both the EG and CG as a pre-test at the beginning of the first week and as a post-test after the tenth week. This design allowed for the assessment of changes in participants’ self-efficacy over the course of the study and the comparison of these changes between the two groups.

4.3.2 Higher-order thinking skill (HOTS) scale

HOTS encompass cognitive abilities that extend beyond basic memorization and comprehension, focusing on three main dimensions: critical thinking, problem-solving, and creativity (Huang et al., 2022; Hwang et al., 2018). To assess these dimensions, this study employed a Chinese translation of the HOTS scale developed by Hwang et al. (2018). The original scale consists of 11 items, with 4 items each for critical thinking and problem-solving, and 3 items for creativity. Each item is rated on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree).

The HOTS scale was translated into Chinese using the back-translation method (Brislin, 1970) to ensure accuracy and clarity. A bilingual expert first translated the original English scale into Chinese, and then another bilingual expert, who had not seen the original English version, translated the Chinese version back into English. Any discrepancies were resolved through discussion and revision of the Chinese translation.

To evaluate the reliability of the translated scale, a pilot study was conducted with 30 undergraduate students not part of the main study. Cronbach’s alpha coefficients were calculated for each dimension of the scale. The results indicated good internal consistency for critical thinking ($\alpha=0.81$), problem-solving ($\alpha=0.84$), and creativity ($\alpha=0.78$), demonstrating sufficient reliability for use in this study (Nunnally, 1978).

The HOTS scale was administered to both the EG and CG as a pre-test at the beginning of the first week and as a post-test after the tenth week. This design allowed for the assessment of changes in participants’ HOTS over the course of the study and the comparison of these changes between the two groups. By employing this validated measurement tool, the study aimed to gain insights into the development of students’ critical thinking, problem-solving, and creativity skills, which are crucial for success in programming education and beyond (Li et al., 2023).

4.3.3 Creative product analysis Matrix(CPAM)

To assess students’ project implementation skills, this study employed the CPAM developed by Besemer and O’Quin (1999). CPAM evaluates creative products across three primary dimensions: Novelty, Resolution, and Elaboration & Synthesis.

Novelty assesses the originality and innovativeness of the product, focusing on new ideas, concepts, or approaches. Resolution evaluates the functionality, effectiveness, and appropriateness of the product in addressing the identified problem or achieving its intended purpose. Elaboration and Synthesis, also known as Style, assesses the complexity, sophistication, and coherence of the product.

In this study, CPAM was used to comprehensively evaluate the projects created by the students. Each of the nine indicators across the three dimensions was assessed using a 5-point Likert scale, with 1 indicating low performance and 5 indicating high performance. Two independent experts in programming education and creativity research scored each student's creative output. The final score for each indicator was determined by averaging the experts' ratings.

To ensure reliability, inter-rater reliability was assessed by computing the intra-class correlation coefficient (ICC) between the experts' scores. The ICC values ranged from 0.75 to 0.86 (Table 1), indicating a high level of agreement between the experts (Koo & Li, 2016). These results demonstrate the reliability of the CPAM assessment in this study.

The application of the CPAM model in this study aligns with its established use in evaluating students' project implementation skills and creative output in educational settings (Hsiao et al., 2023; Lee et al., 2023b). By leveraging this comprehensive framework, the study aims to provide a reliable assessment of students' creative products within the domain of programming education, informing instructional strategies and supporting the development of creative problem-solving skills. The CPAM assessment was conducted during the tenth week of the study, following the project presentations, to evaluate the impact of ChatGPT-generated personalized feedback reflective reports on students' project implementation skills.

4.4 Ethical considerations in AI-Generated feedback

While this study received approval from the Institutional Review Board and maintained participant privacy through data anonymization, several additional ethical considerations specific to AI-generated feedback were carefully addressed throughout the research design and implementation.

First, to mitigate potential bias in ChatGPT-generated explanations, we implemented multiple validation measures. During system development, a panel of three programming education experts reviewed samples of AI-generated feedback for accuracy, clarity, and pedagogical soundness. This expert validation helped identify instances where the AI system might provide misleading or oversimplified explanations of programming errors. Additionally, the system was designed to generate feedback based on established programming principles rather than subjective interpretations, further reducing potential bias. The carefully engineered prompts

Table 1 Scorer reliability of the CPAM (Besemer & O'Quin, 1999)

Subscales	indicators	Scorer reliability
1. Novelty	1.1 Original	0.79*
	1.2 Surprising	0.78*
2. Resolution	2.1 Valuable	0.86*
	2.2 Logical	0.78*
	2.3 Useful	0.75*
	2.4 Understandable	0.76*
3. Elaboration & Synthesis	3.1 Organic	0.75*
	3.2 Elegant	0.82*
	3.3 Well-Crafted	0.79*

* $p < 0.05$

(described in subsection 3.2) were specifically designed to emphasize objective error identification and explanation rather than prescriptive solutions.

Second, we recognized the risk of over-reliance on AI-generated feedback potentially diminishing students' self-directed reflection capabilities. To address this concern, our implementation deliberately positioned ChatGPT as a reflective aid rather than a solution provider. The sequencing of learning activities was crucial—students first engaged independently with programming challenges before receiving AI-generated feedback, thereby preserving the cognitive space necessary for self-directed problem-solving. Moreover, the reflective reports were structured to promote meta-cognitive development by asking guiding questions rather than providing direct solutions, encouraging students to develop their own problem-solving strategies based on the personalized insights.

Third, data privacy and security received particular attention, especially regarding the collection and processing of student error logs. All data collection procedures were transparently communicated to participants during the informed consent process. The Firebase database storing student code and error logs implemented industry-standard encryption and access controls, with data access restricted to authorized researchers only. No personally identifiable information was shared with the ChatGPT system, and student error logs were used exclusively for generating feedback within the context of this study. Students retained ownership of their data and were informed of their right to request deletion of their information at any time. Furthermore, the implementation did not contribute to the training of the ChatGPT model itself, addressing concerns about student data being incorporated into commercial AI systems.

These ethical safeguards were integrated throughout the research design to ensure that the benefits of AI-generated personalized feedback were realized while minimizing potential ethical risks. Regular monitoring of the system's outputs and student interactions throughout the study allowed for ongoing assessment of these ethical considerations and appropriate adjustments when necessary.

5 Results

5.1 The impact of ChatGPT-generated personalized feedback reflective reports on self-efficacy

To investigate the effect of ChatGPT-generated personalized feedback reflective reports on students' self-efficacy, an ANCOVA was conducted with pre-test scores as the covariate. Levene's test was performed to ensure the homogeneity of variances assumption was met ($p > 0.05$), and the results indicated that this assumption was satisfied for all five dimensions of self-efficacy (Table 2).

The self-efficacy scale consisted of five dimensions: Logical Thinking, Cooperation, Algorithm, Control, and Debug. Descriptive statistics for each dimension are presented in Table 3, showing the mean and standard deviation scores for both the EG and CG.

Table 2 The Levene's test results for self-efficacy

Variable	Levene's test	
	F	<i>p</i>
Logical Thinking	2.52	0.116
Cooperation	0.50	0.481
Algorithm	2.84	0.096
Control	3.32	0.072
Debug	1.28	0.261

Table 3 Descriptive results for self-efficacy

	EG (<i>N</i> =40)				CG (<i>N</i> =39)			
	Pre-test		Post-test		Pre-test		Post-test	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Logical Thinking	11.6	1.66	15.7	2.07	11.1	1.70	14.2	1.61
Cooperation	9.05	1.11	10.1	1.28	9.23	1.01	9.79	1.20
Algorithm	8.70	0.91	11.5	1.38	8.59	1.12	10.7	1.14
Control	9.13	0.91	10.6	1.15	8.95	1.00	10.0	1.00
Debug	9.07	0.97	10.8	1.26	9.18	1.12	9.97	1.16

Table 4 ANCOVA results for self-efficacy

Variable	SS	df	Mean Square	F	<i>p</i>	Partial η^2
Logical Thinking	31.5	1	31.52	9.50	0.003**	0.111
Cooperation	2.43	1	2.43	1.65	0.203	0.021
Algorithm	11.48	1	11.48	7.49	0.008**	0.056
Control	3.06	1	3.06	2.62	0.109	0.033
Debug	14.48	1	14.48	10.43	0.002**	0.121

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

The ANCOVA results (Table 4) revealed significant differences between the EG and CG in three dimensions of self-efficacy: Logical Thinking ($F=9.50$, $p=0.003$, partial $\eta^2=0.111$), Algorithm ($F=7.49$, $p=0.008$, partial $\eta^2=0.056$), and Debug ($F=10.43$, $p=0.002$, partial $\eta^2=0.121$). Students in the EG, who received ChatGPT-generated personalized feedback reflective reports, demonstrated higher levels of self-efficacy in these dimensions compared to their counterparts in the CG, who engaged in traditional guided reflection.

In practical terms, these results indicate that students who received personalized feedback from ChatGPT felt significantly more confident in their ability to think logically about programming problems, design effective algorithms, and identify and fix errors in their code. The magnitude of these differences (as shown by the effect sizes) suggests that the ChatGPT-based feedback had the strongest impact on students' debugging confidence (partial $\eta^2 = 0.121$) and logical thinking confidence (partial $\eta^2 = 0.111$), representing meaningful improvements in these crucial programming competencies. These enhanced self-efficacy dimensions are particularly valuable for programming students, as confidence in one's logical reasoning and debugging abilities often translates to greater persistence when facing challenging programming tasks.

To account for potential Type I error inflation due to multiple comparisons across the five self-efficacy dimensions, we applied a Bonferroni correction, adjusting the significance threshold to $p < 0.01$ ($0.05/5$). Even with this more conservative threshold, the differences in Logical Thinking ($p = 0.003$), and Debug ($p = 0.002$) remained statistically significant, while Algorithm ($p = 0.008$) remained marginally significant, providing robust evidence for the effectiveness of the ChatGPT-based approach for these crucial programming competencies.

The personalized feedback provided by ChatGPT, targeting students' specific misconceptions and error patterns, likely contributed to the enhanced self-efficacy in Logical Thinking, Algorithm, and Debug. By receiving tailored guidance and explanations, students in the EG were better equipped to understand and address their challenges, leading to increased confidence in their abilities to think logically, develop algorithms, and debug their code. However, no significant differences were found between the EG and CG in the dimensions of Cooperation and Control. This suggests that ChatGPT-generated personalized feedback reflective reports may have a more pronounced impact on self-efficacy dimensions directly related to programming skills, rather than broader dimensions like Cooperation and Control.

These findings address the first research question, highlighting the potential of ChatGPT-generated personalized feedback reflective reports in enhancing students' self-efficacy in key programming-related dimensions. By providing targeted guidance and support, this innovative approach can help students develop greater confidence in their abilities to think logically, design algorithms, and debug their code, fostering a more positive and self-assured learning experience in programming education.

5.2 The impact of ChatGPT-generated personalized feedback reflective reports on HOTS

This study examined the impact of ChatGPT-generated personalized feedback reflective reports on students' HOTS using an ANCOVA, with pre-test scores as the covariate. Levene's test (Table 5) confirmed that the homogeneity of variances assumption was met for all HOTS dimensions ($p > 0.05$).

Descriptive statistics for HOTS (Table 6) revealed that the EG had higher post-test scores in all dimensions compared to the CG.

The ANCOVA results (Table 7) showed significant differences between the EG and CG in the dimensions of critical thinking ($F = 14.17$, $p < 0.001$, partial $\eta^2 = 0.157$) and problem-solving ($F = 17.96$, $p < 0.001$, partial $\eta^2 = 0.191$). Students in the EG, who received ChatGPT-generated personalized feedback, demonstrated higher levels of critical thinking and problem-solving compared to those in the CG, who engaged in traditional guided reflection.

Table 5 The Levene's test results for HOTS

Variable	Levene's test	
	F	p
Critical thinking	0.297	0.588
Problem-solving	1.55	0.217
Creativity	0.545	0.463

Table 6 Descriptive results for HOTS

	EG (<i>N</i> =40)				CG (<i>N</i> =39)			
	Pre-test		Post-test		Pre-test		Post-test	
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>
Critical thinking	12.5	1.80	16.0	2.20	12.2	1.42	13.9	2.52
Problem-solving	12.3	0.95	15.0	1.67	11.9	1.12	13.7	1.34
Creativity	9.53	1.36	10.4	1.22	9.54	1.05	9.87	1.24

Table 7 ANCOVA results for HOTS

Variable	SS	df	Mean Square	F	<i>p</i>	Partial η^2
Critical thinking	78.2	1	78.17	14.17	<0.001***	0.157
Problem-solving	40.77	1	40.77	17.96	<0.001***	0.191
Creativity	5.51	1	5.51	3.62	0.061	0.045

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

These findings translate to meaningful educational benefits for students. In practical terms, students who received ChatGPT-generated feedback demonstrated substantially enhanced ability to analyze programming problems critically (with scores approximately 2.1 points higher than the CG) and develop effective problem-solving strategies (with scores about 1.3 points higher). The effect sizes (partial $\eta^2 = 0.157$ for critical thinking and 0.191 for problem-solving) indicate that the personalized feedback approach accounted for approximately 16–19% of the variance in these crucial higher-order thinking skills—a substantial educational effect. For programming educators, this suggests that implementing ChatGPT-generated feedback could help students develop the analytical and problem-solving capabilities essential for addressing complex programming challenges beyond the classroom.

To address the potential Type I error inflation due to multiple comparisons across the three HOTS dimensions, we applied a Bonferroni correction by adjusting the significance threshold to $p < 0.017$ ($0.05/3$). Even with this more conservative threshold, the differences in critical thinking ($p < 0.001$) and problem-solving ($p < 0.001$) remained highly significant, providing strong evidence for the effectiveness of the ChatGPT-based approach in enhancing these crucial cognitive skills.

The personalized feedback provided by ChatGPT likely enhanced critical thinking and problem-solving skills by offering targeted guidance based on students' specific misconceptions and error patterns. This tailored approach helped students better understand and address their challenges, fostering the development of these essential HOTS dimensions. However, no significant difference was found between the EG and CG in the creativity dimension. This suggests that ChatGPT-generated personalized feedback reflective reports may have a more pronounced impact on HOTS dimensions directly related to logical reasoning and problem-solving, rather than on creativity.

These findings address the second research question, highlighting the potential of ChatGPT-generated personalized feedback reflective reports in enhancing students' critical thinking and problem-solving skills in programming education. By providing targeted support and guidance, this innovative approach can help students develop

Table 8 Independent samples T-Test results for CPAM model

CPAM	Indicator	Group	M	SD	t-value	<i>p</i>	<i>d</i>
Novelty	Original	EG	3.13	1.04	1.40	0.166	0.315
		CG	2.82	0.89			
	Surprising	EG	3.25	1.03	1.52	0.132	0.342
		CG	2.92	0.87			
Resolution	Valuable	EG	3.98	0.80	2.35	0.022*	0.528
		CG	3.54	0.85			
	Logical	EG	3.55	0.88	2.22	0.029*	0.501
		CG	3.10	0.91			
	Useful	EG	3.63	1.01	2.49	0.015*	0.560
		CG	3.10	0.85			
Elaboration & Synthesis	Understandable	EG	3.77	1.07	2.36	0.021*	0.531
		CG	3.18	1.17			
	Organic	EG	4.05	0.82	2.06	0.043*	0.464
		CG	3.67	0.84			
	Elegant	EG	3.80	1.02	1.38	0.173	0.310
		CG	3.51	0.82			
	Well-Crafted	EG	4.28	0.78	4.40	<0.001***	0.989
		CG	3.51	0.82			

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

essential HOTS, ultimately promoting more effective learning outcomes in critical thinking and problem-solving abilities.

5.3 The impact of ChatGPT-generated personalized feedback reflective reports on project implementation skills

To assess the impact of ChatGPT-generated personalized feedback reflective reports on students' project implementation skills, an independent samples t-test was conducted using the CPAM model. The results (Table 8) revealed significant differences between the EG and CG in five dimensions of the CPAM model: Valuable ($t = 2.35$, $p = 0.022$, $d = 0.528$), Logical ($t = 2.22$, $p = 0.029$, $d = 0.501$), Useful ($t = 2.49$, $p = 0.015$, $d = 0.560$), Understandable ($t = 2.36$, $p = 0.021$, $d = 0.531$), and Well-Crafted ($t = 4.40$, $p < 0.001$, $d = 0.989$). Students in the EG, who received ChatGPT-generated personalized feedback, demonstrated higher performance in these dimensions compared to their counterparts in the CG, who engaged in traditional guided reflection.

From a practical perspective, these results demonstrate that the ChatGPT-facilitated approach significantly enhanced the quality of students' programming projects. The largest difference was observed in how well-crafted the projects were (with an effect size of 0.989), indicating that students receiving personalized feedback created substantially more polished and refined programming solutions. Their projects were also rated as more valuable (by 0.44 points), more logical (by 0.45 points), more useful (by 0.53 points), and more understandable (by 0.59 points) than those created by students in the control group. These improvements represent meaningful enhancements in project quality that would be readily apparent to instructors, potential employers, or users of the software. The effect sizes ranging from medium

(0.501) to large (0.989) indicate that these differences would be noticeable in real-world educational and professional contexts, suggesting that the ChatGPT approach helps students create programming projects that better meet industry standards for quality and usability.

To account for potential Type I error inflation due to multiple comparisons across the nine CPAM indicators, we applied a Bonferroni correction by adjusting the significance threshold to $p < 0.0056$ ($0.05/9$). With this conservative threshold, only the Well-Crafted indicator ($p < 0.001$) remained statistically significant at the adjusted level. However, the consistent pattern of medium to large effect sizes across multiple related indicators (Valuable: $d = 0.528$; Logical: $d = 0.501$; Useful: $d = 0.560$; Understandable: $d = 0.531$; Organic: $d = 0.464$) provides compelling evidence for the practical significance of the ChatGPT intervention on project implementation skills, even if some indicators do not meet the stringent Bonferroni-adjusted significance level.

The personalized feedback provided by ChatGPT likely enhanced project implementation skills by offering targeted guidance and support based on students' specific challenges and error patterns. By addressing individual needs and misconceptions, ChatGPT-generated personalized feedback reflective reports helped students develop more valuable, logical, useful, understandable, and well-crafted projects.

These findings address the third research question, highlighting the potential of ChatGPT-generated personalized feedback reflective reports in improving students' project implementation skills in programming education. By providing tailored guidance and support, this innovative approach can help students create projects that are more valuable, logical, useful, understandable, and well-crafted, ultimately leading to demonstrably improved project implementation outcomes.

6 Discussion

6.1 The impact of ChatGPT-generated personalized feedback reflective reports on self-efficacy

This study demonstrates that ChatGPT-generated personalized feedback reflective reports, grounded in the Cognitive Apprenticeship Model (Collins et al., 1991), significantly enhance students' self-efficacy in Logical Thinking, Algorithm, and Debug dimensions compared to traditional guided reflection. These results align with recent research highlighting the importance of personalized feedback in programming education (Deeva et al., 2021; Qian & Lehman, 2019).

The ChatGPT-generated reports provided targeted feedback based on students' specific error patterns, addressing their unique challenges and misconceptions. This approach, consistent with Zone of Proximal Development theory proposed by Vygotsky and Cole (1978), proved more effective than traditional guided reflection, which often relies on predefined questions that may not capture individual learners' needs (Wu, 2023). Personalized explanations and suggestions enabled students in the experimental group to better understand and overcome their difficulties, boosting their self-efficacy in Logical Thinking, Algorithm, and Debug (Chen et al., 2023; Fukuda et al., 2022).

Moreover, the ChatGPT-based system, informed by Scaffolding theory (Wood et al., 1976), offered actionable recommendations for improvement, empowering students to take concrete steps to enhance their programming skills. Traditional guided reflection often lacks this targeted guidance, leaving students without sufficient support to identify and address their weaknesses (Binks et al., 2022). The personalized feedback acted as scaffolding, helping students develop problem-solving strategies and debugging techniques, thereby fostering their self-efficacy (Moon et al., 2020; Wood et al., 1976).

However, the study found no significant differences between the EG and CG in Cooperation and Control dimensions. This may be due to the ChatGPT-generated reports focusing primarily on individual programming skills, while traditional guided reflection may prompt students to consider aspects related to collaboration and project management. Future iterations of the ChatGPT-based feedback system could integrate guidance on effective teamwork and project planning, drawing from Collaborative Learning theory (Johnson & Johnson, 1999) and Project-Based Learning theory (Blumenfeld et al., 1991), to support students' self-efficacy in these areas (Hsu et al., 2021; Yadav et al., 2021).

These findings contribute to the growing body of literature comparing personalized feedback approaches to traditional guided reflection in programming education. By demonstrating the effectiveness of ChatGPT-generated personalized feedback reflective reports in enhancing students' self-efficacy, particularly in Logical Thinking, Algorithm, and Debug dimensions, this research highlights the potential of large language models to provide targeted support that surpasses the limitations of predefined reflection prompts.

In conclusion, ChatGPT-generated personalized feedback reflective reports, grounded in Cognitive Apprenticeship, Zone of Proximal Development, and Scaffolding theories, significantly enhanced students' self-efficacy in Logical Thinking, Algorithm, and Debug dimensions compared to traditional guided reflection by providing targeted guidance and actionable recommendations. While the feedback system did not significantly impact Cooperation and Control dimensions, future research could explore integrating collaborative learning and project management guidance to further enhance the benefits of personalized feedback over traditional reflection methods. These findings contribute to the ongoing exploration of LLMs in supporting students' learning and self-efficacy development in programming education.

6.2 The impact of ChatGPT-generated personalized feedback reflective reports on HOTS

This study reveals that ChatGPT-generated personalized feedback reflective reports, grounded in Reflective Learning theory (Dewey, 1933; Moon, 2013), significantly improved students' HOTS in critical thinking and problem-solving dimensions compared to traditional guided reflection. These findings align with recent research emphasizing the importance of targeted feedback in promoting HOTS in programming education (Lee et al., 2024; Schnieder & Williams, 2022).

The ChatGPT-based system provided personalized feedback encouraging students to analyze their code, identify errors, and evaluate their problem-solving strategies.

This approach, consistent with Metacognitive theory (Flavell, 1979), proved more effective than traditional guided reflection, which often relies on predefined questions that may not adequately capture individual learners' thought processes and challenges (Sweet, 2022). Engaging in personalized reflection, students in the experimental group developed a deeper understanding of their learning experiences and enhanced their critical thinking abilities (Anggraeni et al., 2023; Pham et al., 2020).

Furthermore, the ChatGPT-generated reports, informed by Problem-Based Learning theory (Barrows, 1996), offered specific insights and recommendations for improving problem-solving strategies based on students' error patterns. Traditional guided reflection often lacks this targeted guidance, leaving students without sufficient support to identify and address ineffective problem-solving approaches (Bernius et al., 2022). The personalized feedback helped students consider alternative solutions and develop more efficient strategies, enhancing their problem-solving skills (Bernius et al., 2022; Karaoglan Yilmaz, 2022).

However, the study found no significant difference between the experimental and control groups in creativity. This finding warrants deeper examination. The lack of improvement in creativity may be attributed to several factors inherent to the ChatGPT-generated feedback approach. First, the ChatGPT system was primarily designed to analyze error codes and provide specific feedback on logical and structural issues, which naturally emphasizes convergent thinking processes that lead to correct solutions rather than divergent thinking necessary for creativity (Cropley, 2006; Wu et al., 2024). The prompts used for generating feedback (as described in subsection 3.2) focused on error explanation, misconception identification, and specific improvement suggestions, without explicit prompts for encouraging novel or alternative approaches that might foster creativity.

Second, the nature of programming error feedback may inadvertently reinforce a more structured and rule-based mindset. As noted by Amini et al. (2024), creativity in programming often requires breaking away from conventional patterns and exploring unconventional solutions. The feedback mechanism employed in this study, while effective for logical reasoning and problem-solving, may have emphasized correctness and efficiency over exploration and originality, potentially constraining creative expression (Lee et al., 2023b).

Third, as suggested by Self-Regulated Learning theory (Zimmerman, 2002), developing creativity often requires specific scaffolding that encourages risk-taking, tolerance for ambiguity, and consideration of multiple perspectives (Stoycheva, 2024). While the ChatGPT-generated feedback effectively supported self-regulation in problem-solving processes, it may not have provided the specific types of encouragement and stimulation needed to enhance creative thinking in programming contexts (Lee et al., 2023b).

These findings contribute to the growing body of literature comparing personalized feedback approaches to traditional guided reflection in promoting HOTS in programming education. By demonstrating the effectiveness of ChatGPT-generated personalized feedback reflective reports in enhancing critical thinking and problem-solving skills, this research highlights the potential of large language models to provide targeted support that surpasses the limitations of predefined reflection prompts.

However, the results also indicate that different approaches may be needed to effectively foster creativity in programming education.

In conclusion, ChatGPT-generated personalized feedback reflective reports, grounded in Reflective Learning, Metacognitive, and Problem-Based Learning theories, significantly enhanced students' HOTS in critical thinking and problem-solving dimensions compared to traditional guided reflection by providing targeted guidance and actionable recommendations. While the feedback system did not significantly impact creativity, future research could explore integrating prompts that specifically stimulate divergent thinking, encourage multiple solution pathways, and provide examples of creative problem-solving to further enhance the benefits of personalized feedback over traditional reflection methods. These findings contribute to the ongoing exploration of large language models in supporting HOTS development in programming education, while also highlighting important areas for improvement in fostering creativity.

6.3 The impact of ChatGPT-generated personalized feedback reflective reports on project implementation skills

This study demonstrates that ChatGPT-generated personalized feedback reflective reports, grounded in Authentic Learning theory (Herrington & Oliver, 2000), significantly enhance students' project implementation skills in the dimensions of Valuable, Logical, Useful, Understandable, Organic, and Well-Crafted compared to traditional guided reflection. These results align with recent research highlighting the importance of targeted feedback in improving project-based learning outcomes in programming education (Hsiao et al., 2023; Lee et al., 2023b).

The ChatGPT-based system provided personalized feedback offering insights and suggestions tailored to students' specific needs and goals. This approach, consistent with Situated Learning theory (Lave & Wenger, 1991), proved more effective than traditional guided reflection, which often relies on generic prompts that may not adequately address individual learners' unique challenges and aspirations (Al Shoul et al., 2024). Receiving targeted guidance, students in the experimental group developed projects with greater perceived value and relevance, aligning with Authentic Learning principles (Herrington & Oliver, 2000; Tsai, 2023).

Moreover, the ChatGPT-generated reports, informed by Cognitive Load theory (Sweller, 1988), provided specific feedback on the structure, organization, and clarity of students' projects. Traditional guided reflection often lacks this targeted guidance, leaving students without sufficient support to improve the logical flow and comprehensibility of their projects (Kukreja & Singh, 2019; Wang et al., 2019). The personalized feedback helped students create projects that were more accessible and engaging, consistent with Cognitive Load theory (Sweller, 1988).

The ChatGPT-based system also offered practical suggestions for enhancing the functionality and applicability of students' projects. This personalized guidance, grounded in Situated Learning theory (Lave & Wenger, 1991), proved more effective than traditional guided reflection, which may not provide adequate support for students to make their projects more useful and relevant to real-world contexts (Barros et al., 2020; O'Dowd, 2023). Receiving targeted recommendations, students in the

experimental group created projects that were more impactful, aligning with Situated Learning theory (Lave & Wenger, 1991).

Finally, the ChatGPT-generated reports, informed by Constructionist Learning theory (Papert & Harel, 1991), helped students create projects that were more cohesive, polished, and aesthetically pleasing, demonstrating higher craftsmanship and attention to detail. Traditional guided reflection often lacks this targeted feedback, leaving students without sufficient support to refine and improve the overall quality of their projects (Wu et al., 2023a). The personalized guidance supported students in the iterative refinement of their projects, consistent with the Constructionist approach (Papert & Harel, 1991).

These results contribute to the growing body of literature comparing personalized feedback approaches to traditional guided reflection in enhancing project implementation skills in programming education. By demonstrating the effectiveness of ChatGPT-generated personalized feedback reflective reports in improving project quality across multiple dimensions, this research highlights the potential of large language models to provide targeted support that surpasses the limitations of predefined reflection prompts.

In conclusion, ChatGPT-generated personalized feedback reflective reports, grounded in Authentic Learning, Situated Learning, Cognitive Load, and Constructionist Learning theories, significantly enhanced students' project implementation skills in the dimensions of Valuable, Logical, Useful, Understandable, Organic, and Well-Crafted compared to traditional guided reflection by providing targeted guidance and actionable recommendations. These findings contribute to the ongoing exploration of LLMs in supporting project-based learning and skill development in programming education, showcasing the advantages of personalized feedback over traditional reflection methods.

7 Conclusions

This study investigated the impact of ChatGPT-generated personalized feedback reflective reports on self-efficacy, HOTS, and project implementation skills in a university-level Python programming course. The results demonstrate that the ChatGPT-based approach, grounded in established learning theories, significantly enhanced undergraduate students' self-efficacy, critical thinking, problem-solving, and project implementation skills compared to traditional guided reflection. The research contributes to the growing literature on integrating ChatGPT in higher education programming courses and highlights the potential of personalized feedback in promoting self-efficacy and HOTS development in university settings. The findings offer valuable insights for educational practitioners and researchers into leveraging AI-powered feedback systems in reflective learning approaches within higher education.

The success of this approach suggests potential applications across various disciplines in higher education where complex problem-solving and critical thinking are equally crucial. The personalized nature of the feedback could address diverse learning needs in university classrooms, potentially enhancing programming skill development as evidenced by the improved project outcomes and higher-order thinking

skills demonstrated in this study. The study underscores the importance of integrating cutting-edge technology into higher education practices. Exposing undergraduate students to AI applications not only enhances their learning experience but also prepares them for future careers in increasingly technology-driven industries. Moreover, the enhanced project implementation skills are particularly relevant to higher education contexts where applying theoretical knowledge to practical projects is paramount and aligns with industry expectations.

While our results demonstrate the immediate benefits of ChatGPT-generated personalized feedback, important questions remain about its long-term implications. A significant consideration is whether reliance on AI-generated feedback might create dependence that could hinder students' independent debugging and problem-solving skill development. Although our study showed improved self-efficacy and HOTS during the course, these gains might not necessarily translate to sustained benefits or transfer to novel programming contexts without AI assistance. Students might develop expectations for immediate, tailored guidance that is unavailable in many professional settings, potentially limiting their capacity for independent problem-solving. Future longitudinal research should examine whether the benefits of AI-assisted reflection persist beyond the immediate course context, how students calibrate between seeking AI assistance and independent problem-solving, and how these interventions might be designed to progressively scaffold students toward greater self-regulation and autonomous learning. Such studies could assess the retention of self-efficacy, HOTS, and problem-solving abilities several months after the intervention to determine whether the approach fosters lasting changes in students' cognitive development and programming capabilities, or whether continued AI scaffolding is necessary to maintain these benefits. Additionally, comparative studies examining different intensities and durations of AI-supported feedback could help identify optimal implementation strategies that balance immediate learning benefits with long-term skill development.

While the ChatGPT-generated personalized feedback significantly improved critical thinking and problem-solving, it did not yield significant improvements in creativity. Future iterations of the system could address this limitation through enhanced prompt engineering that explicitly encourages creative thinking. Specifically, ChatGPT prompts could be redesigned to not only analyze errors but also suggest multiple alternative approaches to solving problems, highlight opportunities for novel implementations, and pose open-ended questions that stimulate divergent thinking. For example, after providing error analysis, the system could include prompts like: "Consider three different ways to approach this problem beyond the conventional solution" or "How might this code be reimaged to create a more elegant or innovative outcome?"

Limitations of the study include the specific focus on Python and Raspberry Pi, which may limit generalizability to other programming languages and platforms used in higher education. The study also did not examine long-term impacts or compare the approach to other personalized feedback methods that may be employed in university settings. Additionally, while our experimental design controlled for several variables by providing identical instructional materials, ensuring equal access to instructor support, and maintaining consistent teaching approaches, we did not spe-

cifically measure student motivation and engagement as variables. Future research should address these limitations by incorporating measures of motivation and engagement to better understand how these factors might mediate the effectiveness of AI-generated feedback.

Another significant limitation is the absence of qualitative data capturing students' subjective experiences with the ChatGPT-generated feedback. While our quantitative measures demonstrated significant improvements in self-efficacy, HOTS, and project implementation skills, we did not systematically collect students' reflections on how they perceived and utilized the AI-generated feedback. Such qualitative insights could have provided valuable context for understanding the mechanisms through which personalized feedback influenced learning outcomes. Future research should incorporate mixed-methods approaches that combine quantitative assessments with qualitative data collection methods such as interviews, focus groups, or open-ended reflective journals to explore students' perceptions of AI-generated feedback, their emotional responses to receiving personalized guidance, and their strategies for incorporating this feedback into their learning processes. This qualitative dimension would enrich our understanding of how students engage with AI-facilitated reflective practices and could inform refinements to the content, timing, and delivery of ChatGPT-generated feedback.

The semester-based assignment of participants to experimental and control groups, rather than randomized assignment, presents another limitation, as potential semester-specific factors could have influenced the results despite our efforts to maintain consistency across semesters. Future studies would benefit from randomized assignment within the same timeframe when logistically feasible. Furthermore, our quasi-experimental design could not completely account for potential differences in group dynamics between the experimental and control cohorts, which might have affected the learning environment and subsequently the results.

Future research should expand on this study's findings in several key directions. First, adaptation of ChatGPT-generated feedback for different levels of programming expertise warrants investigation. While this study focused on undergraduate students with relatively similar programming backgrounds, future research could examine how the feedback system might be calibrated for different proficiency levels—from novice programmers requiring more foundational guidance to advanced learners needing more sophisticated algorithmic insights. This differentiated approach could involve adjusting prompt parameters to vary the depth of explanations, complexity of examples, and level of scaffolding in the reflective guidance. For example, beginners might receive more structured scaffolding with concrete examples and step-by-step guidance, while advanced learners could benefit from more abstract principles, optimization techniques, and creative problem-solving challenges. Such adaptive feedback systems could potentially enhance the effectiveness of programming education across diverse student populations.

Second, integrating AI-driven reflective learning with complementary instructional strategies presents promising opportunities. Future studies could explore synergies between ChatGPT-generated personalized feedback and approaches such as flipped classrooms, where students could engage with AI-generated reflective content before classroom sessions, allowing instructors to focus on addressing specific misconcep-

tions and advanced concepts during face-to-face interactions. Similarly, the integration with collaborative learning models could be explored, potentially using shared AI-generated feedback as a foundation for peer discussions, collaborative debugging sessions, or team-based programming projects. This integration could leverage the strengths of human-to-human interaction while benefiting from the personalization capabilities of AI-driven feedback.

Third, investigating the scalability of this approach to larger classroom settings is crucial for broader implementation. This study's methodology, while effective for 79 students, may require modifications for implementation in larger programming courses commonly found in university settings. Future research should examine technical infrastructure requirements, workflow optimizations, and instructor oversight mechanisms needed to maintain feedback quality and timeliness in large-scale deployments. Studies could also explore how automated feedback monitoring systems might help instructors identify common misconceptions across large student cohorts, enabling more targeted interventions during classroom instruction. Additionally, cost-efficiency analyses comparing various implementation models would provide valuable insights for institutions considering adoption of AI-driven reflective learning approaches.

Future research should also explore the integration of collaborative learning and project management guidance, which are crucial skills in higher education and professional environments. Studies investigating the applicability of this approach to other disciplines and interdisciplinary projects in higher education could provide valuable insights into fostering cross-disciplinary thinking and preparing students for diverse career paths. Additionally, examining the long-term retention of skills and knowledge gained through AI-assisted reflection would help establish the sustained impact of this approach beyond immediate course outcomes.

In conclusion, this study demonstrates the effectiveness of ChatGPT-generated personalized feedback reflective reports in enhancing key skills in university-level programming education. It opens new avenues for innovation in higher education, suggesting ways to leverage AI for more personalized and effective instruction that measurably enhances cognitive skills and implementation abilities across various disciplines. As AI in education advances, this research lays the foundation for exploring personalized feedback approaches that foster students' growth and success in higher education, contributing to the development of a more skilled and adaptable workforce in the rapidly evolving technological landscape. The findings have significant implications for curriculum design, pedagogical approaches, and the integration of AI tools in higher education programming courses, potentially transforming how universities prepare students for the challenges of the digital age.

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Data availability The datasets utilized in this research can be obtained from the corresponding author upon reasonable request.

Declarations

Competing interests The authors declare no competing financial interests related to this work.

Conflict of interest The authors confirm that there are no known conflicts of interest, financial or personal, that could have potentially influenced the outcomes of this research.

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